Epitech AeroTech Industries

Optimizing Predictive Maintenance Algorithms

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Supervisors / Mentors:

- Dr. Alan Turing (Academic Supervisor)
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Abstract

This report details a six-month internship at AeroTech Industries, focusing on the optimization of predictive maintenance algorithms for commercial aircraft. By implementing new Machine Learning techniques, false alerts were reduced by 14%. This document covers the technical context, the methodology used, and the results of the A/B testing phase.

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Part 1: Technical Handover

This section is intended for a newcomer taking over the project. It describes the company context, the project's general architecture, and the work organization.

1. Company Context

AeroTech Industries is a leader in aerospace engineering. The team I joined is responsible for the digital twin platform used to monitor aircraft health in real-time.

2. Project Architecture

The project follows a microservices architecture. Data is ingested via MQTT, processed by a Spark cluster, and stored in a time-series database.

2.1. Anomaly Detection Model

To process time-series data from engine sensors, we opted for an architecture based on Random Forests combined with a Long Short-Term Memory (LSTM) network.

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

def preprocess_sensor_data(df, window_size=50):
    # Remove extreme outliers
    df = df[(np.abs(df.temperature - df.temperature.mean()) <= (3 *
df.temperature.std()))]

    # Normalize data
    scaler = StandardScaler()
    df['temp_scaled'] = scaler.fit_transform(df[['temperature']])

    # Create sliding windows for LSTM
    return df.rolling(window=window_size).mean()
```

2.2. Key Terms

LSTM Long Short-Term Memory — a recurrent neural network architecture well-suited for time-series forecasting.

MQTT Message Queuing Telemetry Transport — a lightweight publish-subscribe messaging protocol for IoT devices.

ROC AUC Receiver Operating Characteristic — Area Under Curve, a metric evaluating classifier performance across thresholds.

Digital Twin A virtual replica of a physical system used for real-time monitoring and simulation.

3. Methodology and Implementation

3.1. Data Flow Diagram

The diagram below illustrates the data flow into the hybrid LSTM model.

ARCHITECTURE DIAGRAM PLACEHOLDER
Replace this block with:
`image("path/to/image.png")`

Listing 1: Architecture diagram of the hybrid LSTM network developed for time-series analysis.

4. Results Analysis

4.1. Performance Metrics

Approach	Accuracy	Gains vs. Baseline
Static Rules	78.2%	0%
Random Forest	89.1%	+10.9 pts
Our Model	94.5%	+16.3 pts

Table 1: Summary of performance metrics on the test set.

Part 2: Self-Assessment

This section presents an argument to my supervisor regarding my performance, highlighting my strengths and areas for improvement.

5. Strengths

- **Technical Adaptability:** I was able to quickly master the complex Spark infrastructure used at AeroTech.
- **Problem Solving:** I successfully identified a bottleneck in the data ingestion pipeline that was causing 5% data loss.

6. Areas for Improvement

- **Communication:** I need to improve my ability to explain technical concepts to non-technical stakeholders.
- **Time Management:** During the first month, I struggled with prioritizing tasks, though this improved significantly later.

Part 3: Dream Project Pitch

This section is designed to convince senior management to entrust me with a high-impact project.

7. Context

With the rise of autonomous flight systems, there is a growing need for edge-computing anomaly detection.

8. The Project: "EdgeGuard"

I propose developing a lightweight version of our LSTM model that can run directly on the aircraft's hardware (Edge AI), reducing latency and satellite bandwidth costs.

9. Interest for the Company

Implementing EdgeGuard would reduce operational costs by 20% and provide real-time safety critical alerts without relying on ground connectivity.

10. Conclusion

This internship allowed me to apply theoretical Machine Learning concepts to critical industrial problems. The results obtained pave the way for a generalization of this approach to other aircraft subsystems.